SIADS 696 Milestone II Project Report

Neurodiversity and Archetypes: Predicting ADHD Signals and Discovering Jungian-like Personality Clusters via Big Five–to–MBTI Feature Engineering

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1. Introduction

The project aims to explore neurodiversity in the workplace and personality structure through two complementary machine learning analyses, addressing two problems. First, workplace neurodiversity, particularly ADHD, is often underrecognized and poorly understood in organizational settings. We aim to determine whether self-reported ADHD signals among tech workers can be predicted using workplace environment, demographic characteristics, and contextual economic features in a pooled multi-year survey dataset collected from tech workers. The impact of this analysis is that it can guide evidence-based organizational policies, helping employers design work environments that better support neurodiverse employees. Second, personality research itself faces a structural problem. Trait models like the Big Five are dimensional and statistically robust, while typological systems like Jungian or MBTI-style frameworks are widely used but often criticized for oversimplification. We therefore ask whether stable, reproducible clusters can be discovered when Big Five traits are mapped to continuous Jungian-style proxies, and whether some clusters display patterns frequently discussed in ADHD literature, such as relatively higher Openness and lower Conscientiousness.

Together, the broader impact is a comprehensive understanding of neurodiversity that integrates trait structure and environmental context, supporting both organizational insight and theoretical clarity. We were motivated to pursue this project by the growing public conversation around neurodiversity in the tech industry and the gap between popular personality narratives and rigorous data-driven evidence.

Our project combines supervised (Part A) and unsupervised methods (Part B) to study neurodiversity from predictive and structural perspectives. In Part A we train a classifier on OSMI tech worker survey data, addressing high dimensionality and class imbalance. In Part B we perform stability-validated clustering on Big Five traits augmented with engineered proxies to discover interpretable personality structures without imposing typological labels. Our novel contribution is linking workplace-based ADHD signal prediction with trait-space archetype discovery in a unified framework.

Despite many prior projects that focus on text-based classification from social media posts, our project predicts self-reported ADHD signals using structured workplace survey data augmented with contextual economic features, shifting the focus from linguistic detection to environment-versus-trait interaction. In addition, we complement supervised prediction with unsupervised personality structure discovery.

1.1 Part A Findings

- **Best model:** Decision Tree (max_depth=5, min_samples_leaf=15, max_leaf_nodes=10) achieved CV recall 0.827 ± 0.036, a 13.6% improvement over the untuned baseline (0.728).
- **Imbalance handling:** Customised class weight (~1:6, EDA-derived) outperformed SMOTE on CV recall and was selected for its transparent, reproducible derivation from the observed class ratio.
- **Primary signal:** Comorbidity features dominate — addictive disorder (importance 0.302), continent_work (0.273), willing_share_family (0.136), mood (0.121). Ablation confirms comorbidity removal causes the largest recall drop (−0.091).
- **Geographic dependence:** Geographic ablation causes a −0.076 recall drop and nearly doubles prediction instability (std 0.043 → 0.081), raising deployment fairness concerns for non-American populations.
- **Data-limited:** Learning curve shows CV recall still rising at full training size. Train-validation gap of 0.077 indicates moderate overfitting; additional ADHD-positive responses would improve performance.

1.2 Part B Findings

- **Primary configuration:** K-Means (K=2) with engineered features achieved Silhouette 0.200 and near-perfect seed stability (ARI = 0.996).

- **Cluster structure:** Two archetype-like profiles emerged — one with higher Openness, Agreeableness, and J_proxy with lower Impulsivity_Index; the other with lower trait scores, elevated Impulsivity_Index, and Creative_Divergence.
- **Robustness:** Subsample stability (mean ARI = 0.970 across 12 subsamples) and proxy-weighting sensitivity (ARI range 0.838–1.000) confirm the partition is not an artefact of initialisation or proxy formula choices.
- **Alternative structure:** K-Means (K=3) is the strongest non-trivial alternative; GMM (K=3, diagonal covariance) provides consistent soft-assignment refinement suggesting meaningful sub-structure.

2. Related Work

A Classified ADHD Dataset based on Ensemble Machine Learning from Social Media: A related study built a binary ADHD text dataset from Reddit posts using expert labelling, staged annotation, and an ensemble of five classifiers, achieving ~95% accuracy on ~18,000 instances with additional LSTM and BERT evaluations (Mostafij Mumu et al., 2022). In contrast, our project uses structured workplace survey data rather than social media text, incorporates workplace environment and geographic context, and complements supervised prediction with unsupervised personality clustering.

Wearable-based ADHD prediction in children: Machine learning models using circadian rhythm features from wearables predict ADHD with AUC up to ~0.798 in a large paediatric cohort (Kim et al., 2023). In contrast, we focus on self-report and contextual features from adult tech professionals, complementing prediction with personality structure discovery rather than physiological or sensor data.

MNAR Sentinel Encoding with Missingness Indicators: A benchmark of missing-value strategies across health and survey databases found that adding missingness indicator variables consistently improved prediction accuracy, confirming that the missingness pattern itself carries predictive signal that imputation alone cannot recover, and that tree-based models handled this well (Grinsztajn et al., 2022). Our project extends this to a longitudinal multi-year survey with structural missingness from survey design changes, requiring sentinel encoding (−1) preserving a three-way distinction between 'not asked', 'answered No', and 'answered Yes'.

Data-driven personality types: Gerlach et al. (2018) applied GMMs to large Big Five samples and identified four reproducible personality types, showing that meaningful cluster structure exists even with modest separation metrics. Our Part B follows this spirit but adds engineered MBTI-style proxy features for interpretability and conducts comprehensive stability validation.

Big Five to MBTI connections: McCrae & Costa (1989) documented systematic correspondences between MBTI dimensions and Big Five traits. Our contribution is to operationalise these as continuous engineered proxies validated via correlation diagnostics and sensitivity checks before clustering.

3. Data Source(s)

3.1 Part A: OSMI Mental Health in Tech Survey

Source: https://osmhhelp.org/research.html. The OSMI Mental Health Tech Workers Survey (2014–2023) collects 60–120 mental-health-related questions per year from employees of tech companies, covering workplace MH policies, disclosure behaviour, comorbidities, demographics, and geography. After excluding 2014 (structural missingness incompatibility) and applying quality filters, the pooled dataset contains approximately 3400 records across nine years with ~15% ADHD-positive class balance. The binary target adhd_signal was constructed from self-reported ADHD diagnosis mentions, explicitly treated as a survey-derived signal rather than a clinical diagnosis.

3.2 Part B: Big Five Personality Test (OpenPsychometrics)

Source: http://openpsychometrics.org/_rawdata/. The dataset contains item-level responses (50 items, Likert 1–5) from 19,719 respondents. OCEAN trait scores were computed by averaging items after reverse-coding per the codebook. Quality control (straight-lining ≥0.90, reverse-consistency mismatch >2.2) removed 165 rows (0.84%), yielding 19,554 records. All five trait scores are z-score standardized prior to feature engineering and clustering to ensure distance-based methods are not biased by raw scale differences.

4. Feature engineering

4.1 Part A Feature Engineering

Preprocessing

1. **Multi-year merge and regex header cleaning:** Column headers across nine survey CSVs contained metadata remnants. A regex-based cleaning function produced normalised column names enabling a multi-year merge.
2. **Column and response harmonisation:** FuzzyWuzzy fuzzy string matching grouped semantically equivalent column names and response values programmatically, with each grouping verified qualitatively before merging to prevent incorrect semantic conflation.
3. **Age outlier removal:** Implausible age values (very large numbers, negatives) were clipped to 16–100, removing biologically impossible entries while preserving the full plausible distribution.
4. **Missing value strategy — MNAR encoding:** Two structurally different missingness types were handled separately. (a) MNAR (structurally absent): columns where a question was not asked in a given year — identified by a year-stratified function with ≥90% missing threshold — were encoded with a −1 sentinel (numeric) or 'Not_Asked' (categorical), plus a companion binary missingness indicator flag per column. (b) MAR/MCAR (randomly missing): remaining missing values received iterative imputation within the ColumnTransformer pipeline. [1]

Feature engineering

1. **Missingness indicators as features:** Binary flags (col_missing=1 if not asked) were added to the feature set so the model can separately learn the predictive effect of belonging to the 'not asked' cohort. Empirical verification confirmed distinct ADHD rates across groups. [13]
2. **Race recoding:** Free-text race responses were mapped to seven standardised demographic categories using regex classification, reducing cardinality while preserving meaningful variation.
3. **Geographic abstraction:** Countries were mapped to six continental regions and US states to four Census regions. This reduces sparsity while retaining the geographic signal.
4. **Job role binarisation:** The multi-label free-text job_spec field was processed via MultiLabelBinarizer, creating binary indicator columns per unique role to enable the model to learn from each job function independently.
5. **Target variable construction:** The binary target adhd_signal was constructed from self-reported ADHD diagnosis questions across survey years. Columns used in construction were added to the leakage exclusion list and removed before any model training.

4.2 Part B Feature Engineering

We construct two representations for clustering: (a) Big Five only (O, C, E, A, N), and (b) Big Five plus engineered continuous proxies intended to be MBTI-style axes and interpretable composites. Importantly, we do not assign discrete MBTI “types”; proxies remain continuous (z-scored). The theoretical motivation for these proxies draws on McCrae & Costa’s (1989) documented correspondences between Big Five traits and MBTI dimensions — further validated by Furnham (1996), who quantified the overlap between MBTI scales and NEO-PI factors — operationalized as weighted linear combinations rather than categorical assignments.

Feature	Definition	Interpretation
N_proxy	$0.7 \times z_O + 0.3 \times z_E$	Intuition-like tendency
J_proxy	$0.8 \times z_C + 0.2 \times z_A$	Judging-like tendency

T_proxy	$0.6 \times z_C - 0.4 \times z_A$	Thinking-like tendency
Impulsivity_Index	$-z_C + z_N$	Low Conscientiousness + high Neuroticism
Emotional_Reactivity	$z_N - 0.3 \times z_A$	Emotional volatility tempered by agreeableness
Creative_Divergence	$z_O - 0.2 \times z_C$	Openness unconstrained by conscientiousness
Social_Adaptability	$z_E + 0.5 \times z_A - 0.3 \times z_N$	Socially adaptive extraversion

Table 1: Part B Engineered Feature Definitions

Sanity checks: we (i) report correlations between proxies and source traits to verify directionality, (ii) test alternative proxy weightings (±30% perturbation) to confirm structural robustness, and (iii) compare cluster solutions with vs. without proxies (ablation) to verify proxies contribute meaningful structure beyond the raw Big Five dimensions.

5. Part A – Supervised Learning

5.1 Methods and Evaluation

This project predicts self-reported ADHD signals among tech workers as a binary supervised classification problem on an imbalanced dataset. The design follows two sequential phases: Phase 1 identifies the best imbalance handling method with a fixed algorithm; Phase 2 compares three mechanistically diverse algorithm families with the selected method fixed. The best Phase 2 model then undergoes sensitivity analysis, learning curve evaluation, feature importance, ablation, and failure analysis on the tuned configuration.

5.1.1 Evaluation metrics

The ADHD signal is imbalanced (~15% positive cases). This project targets the screening phase — minimising false negatives (missed ADHD cases delay support) takes priority over minimising false positives (resolved by further clinical assessment). Recall is therefore prioritised as the primary per-class metric. PR-AUC replaces ROC-AUC as the threshold-independent summary metric: on imbalanced data, ROC-AUC is optimistic because the large negative class makes the false positive rate appear artificially low, while PR-AUC directly reflects performance on the minority class.

5.1.2 Phase 1: Identifying the Best Imbalance Handling Method

Logistic Regression (L1 penalty, C=1, liblinear solver) was fixed to test five imbalance conditions: (a) no correction, (b) sklearn balanced weight, (c) customised EDA-derived weight, (d) SMOTE k=5, and (e) combined SMOTE + balanced. Sklearn balanced and customised weight nearly tied on CV recall (0.680 vs 0.683), but customised weight is selected because it is directly motivated by the observed EDA class ratio (~1:6), providing a transparent and reproducible derivation — whereas sklearn balanced computes weights from each fold's training distribution, introducing slight fold-to-fold variation. Table 2 shows full results.

Method	Parameter	Solution Type	CV recall	Std Dev	F1 score	PR-AUC
Baseline	N/A	Control	0.167	0.053	0.236	0.380
Sklearn	Balanced	Algorithm based	0.680	0.041	0.379	0.382
Customized weight	Derived from EDA	Algorithm based	0.683	0.038	0.376	0.380
SMOTE	K_neighbours = 5	Data based	0.645	0.043	0.376	0.382
Combined (sklearn & SMOTE)	Balanced + Data modification	Algorithm + Data	0.645	0.043	0.376	0.382

Table 2: Phase 1 - imbalance method comparison with Logistic Regression (L1, C=1) fixed as the algorithm. CV recall computed using 5-fold stratified cross-validation on the training set along with F1 and PR-AUC test set.

5.1.3 Phases 2: Algorithm comparison

With customised weight (~1:6) fixed, three mechanistically diverse algorithm families were compared: Bernoulli Naive Bayes (probabilistic, generative), Logistic Regression with L1 regularisation (discriminative, linear), and Decision Tree with pre-pruning (non-parametric, tree-based). Decision Tree achieves the highest CV recall (0.723 ± 0.132) and is selected. It is also chosen for direct interpretability: both the tree structure and SHAP values provide transparent explanations of individual predictions, which is essential for a sensitive application like ADHD screening. Table 3 shows full results.

Model	Mechanism	Weight	CV recall	Standard deviation	F1	PR-AUC
Bernoulli NB (baseline)	Non-probablistic	Customised 1:8	0.510	0.056	0.342	0.312
Logistic Regression	Linear discriminative	Customised 1:8	0.683	0.038	0.378	0.383
Decision Tree	Tree-based	Customised 1:8	0.723	0.132	0.404	0.370

Table 3: Phase 2 – algorithm comparison with customised weight fixed. CV recall computed using 5-fold stratified cross-validation on the training set along with F1 and PR-AUC test set.

5.2 Advanced Model Evaluation & Failure Analysis

5.2.1 Sensitivity Analysis - Decision Tree Pre-Pruning

The untuned Decision Tree (max_depth=5, min_samples_leaf=25, max_leaf_nodes=None) served as baseline (CV recall 0.723 ± 0.132). Three pre-pruning parameters were swept independently using an OFAT (appendix B) (one-factor-at-a-time) design. max_leaf_nodes=10 produces the largest single-parameter gain (0.822 ± 0.028), constraining the tree to 10 leaf nodes and dramatically reducing instability. min_samples_leaf=15 alone gives a modest gain (0.740 ± 0.128). The combined configuration achieves CV recall 0.827 ± 0.036, a 13.6% improvement over baseline. The OFAT design cannot detect parameter interactions — a full factorial search would be a priority extension. Table 3 shows all configurations.

Tuning	max_depth	min samples leaf	max leaf nodes	CV Recall	Standard deviation	Improve (%)
No (baseline)	5	25	None	0.7278	0.1223	N/A
max leaf nodes	5	25	10	0.8223	0.0282	12.98%
min samples leaf	5	15	None	0.7405	0.1278	1.17%
Both	5	15	10	0.8273	0.0359	13.67%

Table 4: Sensitivity analysis – OFAT pre-pruning parameter tuning for Decision Tree. CV recall computed using 5-fold stratified cross-validation.

5.2.2 Learning Curve

The learning curve (Figure 1) plots training recall and CV recall against training set size across 10 intervals (10%–100%), evaluated using 5-fold cross-validation. The train-validation gap of 0.077 at full training size indicates moderate overfitting, partially mitigated by the tuned constraints (max_depth=5, min_samples_leaf=15, max_leaf_nodes=10). Critically, CV recall is still rising at full training size without plateauing, confirming the model is data-limited rather than signal-limited: ~520 ADHD-positive training cases are insufficient for the tree to learn stable splits across all ADHD subgroup profiles. The wide confidence bands at small training sizes reflect cross-validation fold instability driven by class imbalance, addressed only partially by the customised class weight. Additional ADHD-positive responses would directly reduce both the overfitting gap and the data-limited finding.

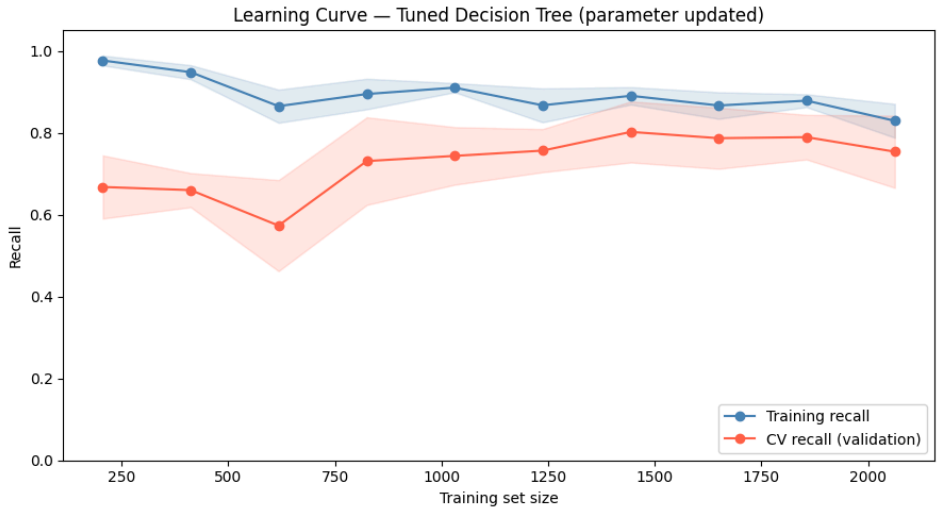

Figure 1: Learning curve for phase 2 tuned decision tree, tested with CV 5 fold recall score, as color bands for standard deviation

5.2.3 Feature importance and ablation analysis

Feature importance was computed using Gini impurity reduction on the tuned Decision Tree. Features are grouped into three interpretive categories.

Comorbidity signal (clinically plausible): addictive disorder (0.269) is the top feature, consistent with the well-established comorbidity between ADHD and addictive disorders through shared impulsivity mechanisms. mood (0.122), mh_treatment_history (0.050) and family_history_mh (0.042) form a coherent clinical cluster representing genuine individual-level ADHD signals.

Geographic confounds: continent_work (0.273) is the second most important feature, which might reflect real regional variation in ADHD diagnosis rates. When continent_work was removed in sensitivity testing, continent_live immediately replaced it, confirming the geographic signal is distributed across both columns — which is why the ablation removes both together.

Workplace disclosure: willing_share_family (0.136), observed_coworker_discuss_mh (0.036), open_to_discuss_coworkers (0.023) and company_size (0.016) reflects comfort disclosing mental health conditions and is partially mediated through diagnosis status.

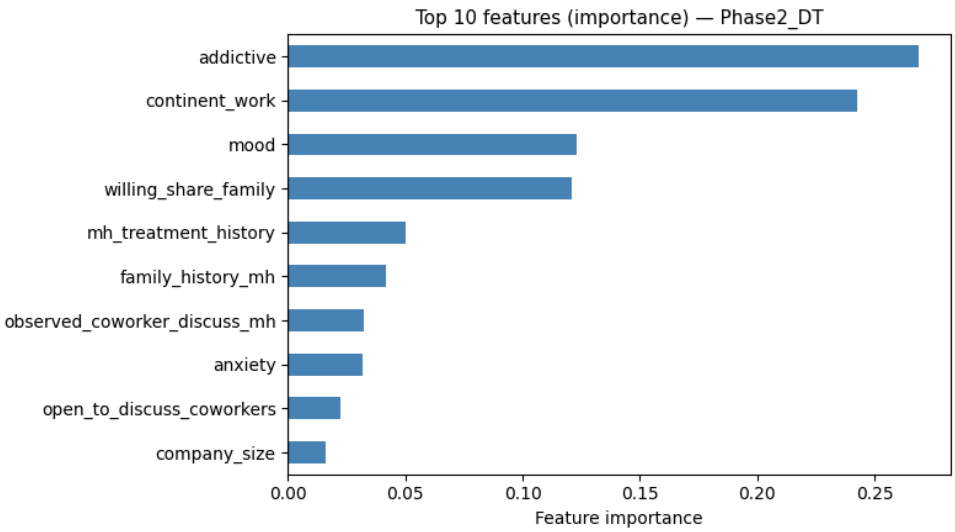

Figure 2: Top 10 features from phase 2 decision tree classifier after sensitivity analysis

5.2.4 Ablation Study

Three feature groups were removed independently and the tuned Decision Tree was retrained under identical 5-fold CV. Table 5 presents results.

Ablated feature	Ablated Features	CV Recall score	Std Dev	Δ Recall
No ablation	None	0.822	0.043	—
Geographic features	continent_work, continent_live	0.746	0.081	-0.076
Workplace disclosure features	willing_share_family, observed_coworker_discuss_mh, open_to_discuss_coworkers	0.758	0.137	-0.064
Comorbidity features	addictive, mood, anxiety, ptsd, ocd, eating, dissociative, etc	0.677	0.179	-0.145

Table 5: Ablation study – three feature groups were removed independently and the tuned decision tree retrained after identical 5-fold CV.

Comorbidity removal causes the largest drop (−0.145), confirming the model is fundamentally a comorbidity screener. Geographic removal causes a near-equal drop (−0.076) while nearly doubling std (0.043 → 0.081) suggest

geographic features were stabilising decision boundaries as well as providing signal, quantifying the fairness concern for non-American populations. Workplace disclosure causes the smallest drop (−0.064) however its std increases (0.043 → 0.137), indicating marginal additive signal.

5.2.5 Failure analysis

SHAP methodology: TreeExplainer decomposes each prediction into additive feature contributions relative to the population baseline E[f(x)]=0.5. Positive SHAP pushes toward ADHD; negative pushes away. Contributions sum exactly to f(x)−E[f(x)]

Instance 1 — False Negative (Geographic Bias): True=1, Pred=0, P(ADHD)=0.104. A genuine ADHD case from a non-American continent with no comorbidities. SHAP shows continent_work (SHAP=−0.24) as the dominant driver — the geographic prior is so strong that this respondent has no viable path to a positive prediction. Root cause: systematic geographic bias. Future fix: region-stratified calibration, fairness-aware training, additional non-American ADHD-positive data collection.

Instance 2 — False Positive (Comorbidity Proxy Conflation): True=0, Pred=1, P(ADHD)=0.628. A non-ADHD American respondent with mood disorder. The model correctly identifies continent=American + mood as high ADHD probability at the population level, but cannot distinguish 'mood with ADHD' from 'mood without ADHD' within a single leaf. Root cause: max_leaf_nodes=10 forces identical predictions for all respondents in the same leaf. Future fix: additional discriminative features (ADHD symptom scales, age of onset), probability calibration.

Instance 3 — Concept Drift (Geographic confound): When 2014 data was included, continent_work became the top feature (Gini 0.465) because 2014 data did not contain information on self reported ADHD signals. Resolution: 2014 excluded entirely. After correction continent_work reduced from 0.465 to 0.273 and addictive disorder correctly emerged as dominant. Future fix: decompose MNAR columns into original value + explicit year indicator.

Note: Refer instance 2 to 3 to appendix B

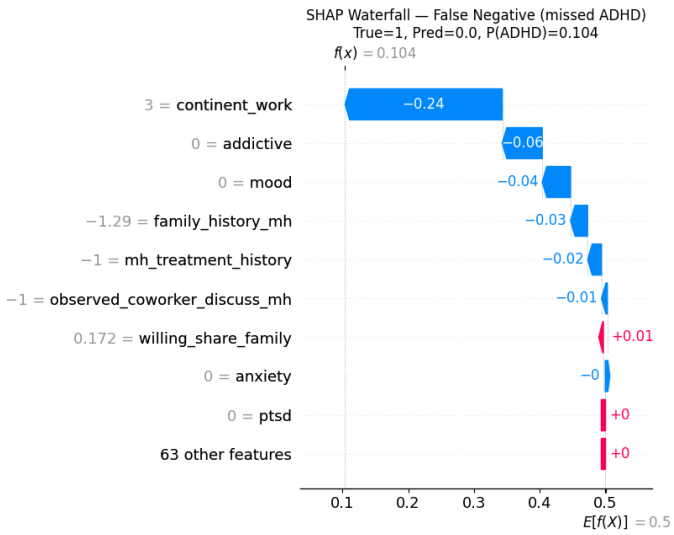

Figure 3: SHAP waterfall plot for instance 1 – False Negative (missed ADHD)

5.3 Key Tradeoffs

- **Recall vs Precision:** The 1:6 custom weight and recall-first CV metric operationalise the screening priority. The tuned DT achieves test recall 0.742 with precision 0.284 — approximately 3 false alarms per true detection at the 0.5 threshold, appropriate for screening where downstream clinical assessment filters false positives.
- **Training data size vs performance:** The learning curve shows CV recall still rising at full training size (train-val gap 0.077). The model is data-limited — additional ADHD-positive survey responses would improve recall without requiring architectural changes.
- **Model complexity vs prediction granularity:** max_leaf_nodes=10 improved CV recall from 0.728 to 0.827 (+13.6%) but limits the model to 10 discrete prediction values, as observed in failure analysis Instance 4. This is a fundamental tradeoff of pruned trees between generalisation and resolution.
- **Geographic signal vs deployment fairness:** Retaining geographic features maximises recall on the current dataset (−0.076 when removed), but the metrics were computed on ~60% American respondents. A model deployed in non-American contexts would systematically miss more ADHD cases than the headline metrics suggest.

6. Part B — Unsupervised Learning

6.1 Research Goal

We investigate whether stable and interpretable archetype-like clusters emerge from Big Five personality traits, and whether augmenting the trait space with continuous Jungian/MBTI-style proxy features yields clearer cluster structure or more psychologically interpretable profiles. This question is motivated by the tension between dimensional trait models (which are statistically robust but lack intuitive typological labels) and categorical personality systems (which are widely understood but psychometrically problematic). Following Gerlach et al. (2018), who demonstrated that Gaussian mixture models can recover reproducible personality types from large Big Five samples, we extend this paradigm by introducing theory-driven engineered features that bridge dimensional and typological perspectives.

6.2 Methods

The Part B pipeline proceeds in ten steps: (1) load raw Big Five item responses; (2) reverse-code designated items per the codebook; (3) compute OCEAN trait means; (4) flag and remove low-quality rows (straight-lining, reverse-consistency); (5) z-score standardize all traits; (6) engineer MBTI-style proxy features; (7) run correlation diagnostics on proxies; (8) grid-search K-Means and GMM across K = 2–8 and two feature sets; (9) evaluate via Silhouette, Davies–Bouldin, and Calinski–Harabasz; (10) perform stability and sensitivity checks.

We fit two unsupervised model families with substantially different distributional assumptions, providing complementary perspectives on cluster structure:

K-Means: a hard-assignment algorithm that minimizes within-cluster sum of squares. We choose K-Means for its computational efficiency and interpretable centroids. The spherical-cluster assumption is a reasonable starting point for z-scored continuous traits. We sweep K from 2 to 8 with n_init = 12 restarts per K, comparing two feature sets (Big Five only vs. Big Five + engineered proxies).

Gaussian Mixture Models (GMM): a probabilistic model that performs soft assignment via the EM algorithm. GMM relaxes the spherical-cluster assumption by fitting per-component covariance matrices (full or diagonal), allowing ellipsoidal cluster shapes that better match the potentially correlated nature of personality traits. We sweep K from 2 to 8 with n_init = 3, testing both full and diagonal covariance types. Soft assignments are particularly appropriate for personality data, where individuals rarely belong exclusively to a single type.

For visualization, we project standardized features to 2D via PCA and plot points colored by cluster assignment. We emphasize that PCA is used strictly as a visualization aid; cluster quality is evaluated via formal internal metrics, not visual separability in a reduced-dimensional projection.

6.3 Model Selection and Internal Evaluation

We evaluate candidate configurations using three complementary internal validity metrics: Silhouette score (higher is better; measures cohesion vs. separation), Davies–Bouldin index (lower is better; penalizes large, close clusters), and Calinski–Harabasz index (higher is better; ratio of between- to within-cluster dispersion). To keep selection

transparent, we report both family-level choices (for interpretability) and cross-method composite ranking (for structural validity). For the main narrative, we use K-Means K=2 as the primary hard-clustering model; as a finer-grained alternative, cross-method rank-sum favors K-Means K=3 on engineered features. Selection priority in this report is explicit: first, select the K-Means family solution for the primary narrative split; second, use cross-method ranking as fine-grained validation and refinement.

Overall Silhouette values are modest (≈0.20), which warrants careful interpretation. In personality science, such values are expected and consistent with the dimensional nature of trait variation: individuals occupy a continuum rather than falling into discrete categories. Gerlach et al. (2018) similarly reported moderate cluster separation in large-sample Big Five analyses while still identifying reproducible types. We therefore treat interpretability and stability as primary decision criteria alongside metric rankings.

Configuration	**K**	**Silhouette**	**Davies-Bouldin**	**Calinski-Harabasz**
K-Means (engineered)	2	0.200	1.754	5578
K-Means (engineered)	3	0.186	1.598	5115
GMM-diag (engineered)	3	0.175	1.638	4787
GMM-diag (engineered)	2	0.188	1.880	4870

Table 6. Top clustering configurations (engineered features). ✓ = primary selected configuration.

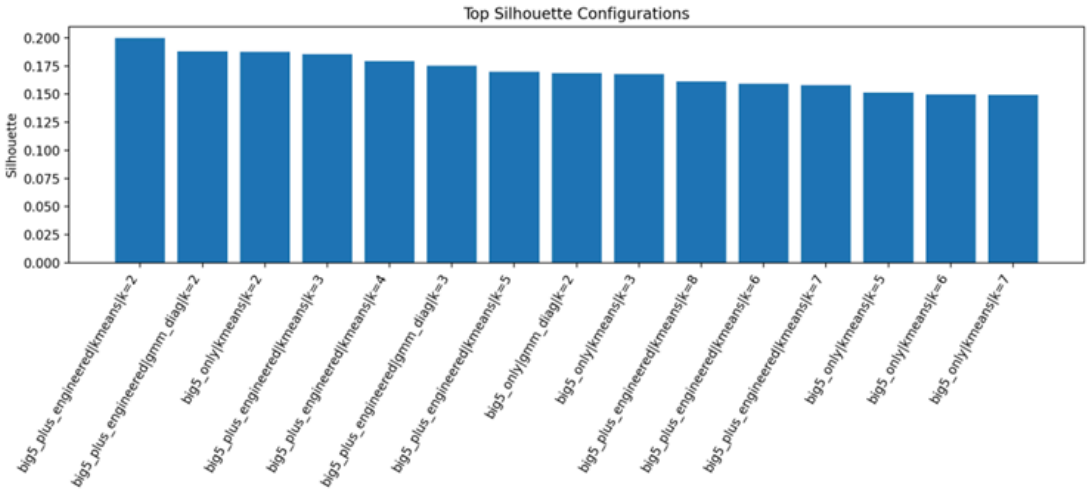

Figure 4 Top Silhouette configurations across model/feature settings.

6.4 Cluster Interpretations

6.4.1 K-Means (K=2) — Engineered Feature Set

K-Means with K=2 produces two broad archetype-like profiles distinguishable along multiple trait and proxy dimensions. Cluster 1 shows higher average Big Five trait scores (notably Openness and Agreeableness) coupled with higher J_proxy (Judging-like structure) and lower Impulsivity_Index, suggesting a profile of psychologically structured, conscientious individuals. Cluster 0 shows comparatively lower trait averages, higher Impulsivity_Index, and elevated Creative_Divergence, consistent with a more impulsive, exploration-oriented profile. These profiles are psychologically coherent and align with the Gerlach et al. (2018) finding that personality structure separates along an overall trait-level axis, while our engineered proxies add interpretive nuance that raw Big Five scores alone cannot provide.

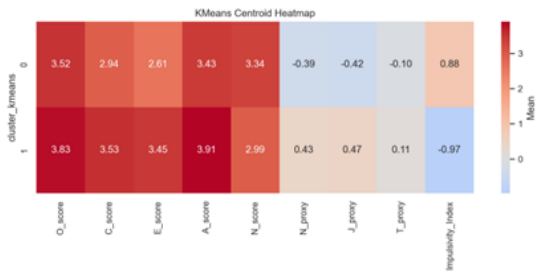

Figure 5. K-Means centroid heatmap

Figure 6. K-Means cluster profile bar chart

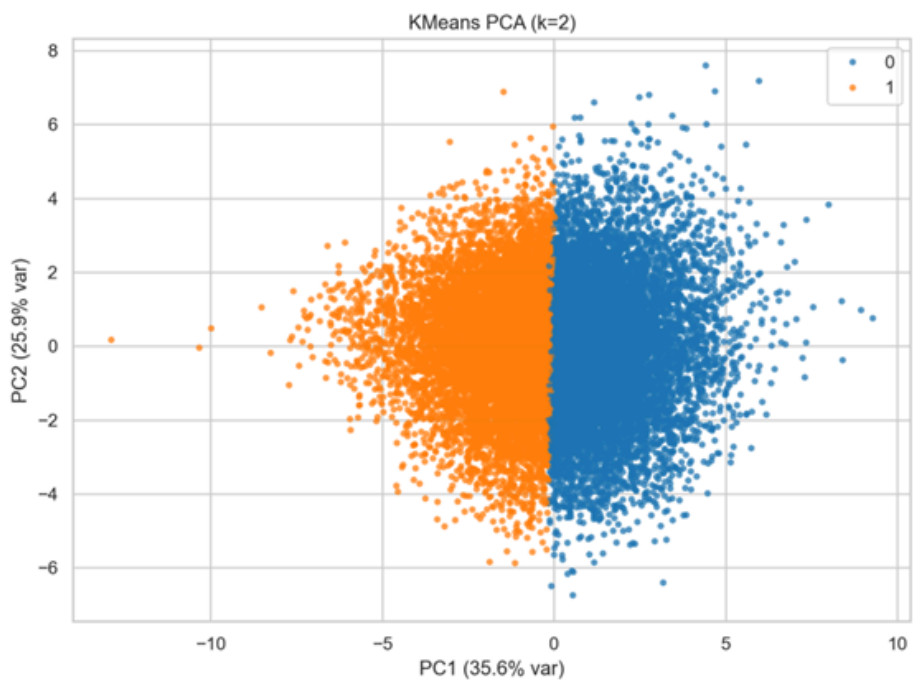

Figure 7. K-Means clusters in PCA space

6.4.2 GMM (K=3, Diagonal Covariance) — Engineered Feature Set

GMM with three components yields a finer-grained partition and provides soft assignments consistent with personality as a continuum. The three components differ meaningfully: Component A displays the lowest Impulsivity_Index with highest J_proxy, representing the most structured profile. Components B and C both show elevated impulsivity, but B is distinguished by higher T_proxy (Thinking-like tendency) while C shows higher

N_proxy (Intuition-like tendency) and Creative_Divergence. This three-way partition suggests that what appears as a single "impulsive" cluster under K-Means is itself heterogeneous, containing distinct sub-profiles that differ in their cognitive style orientation. The soft assignment probabilities confirm that boundary individuals often have substantial posterior probability mass across two components, supporting the psychological position that personality types represent distributional modes rather than discrete categories.

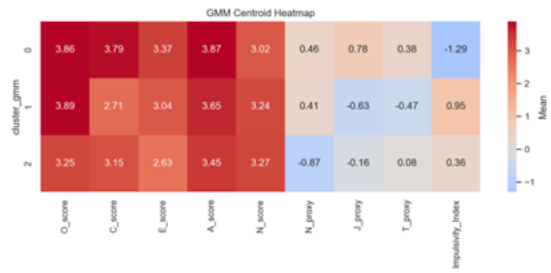

Figure 8. GMM centroid heatmap

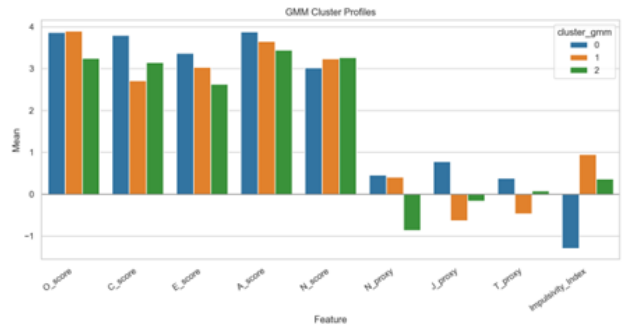

Figure 9. GMM component profile bar chart

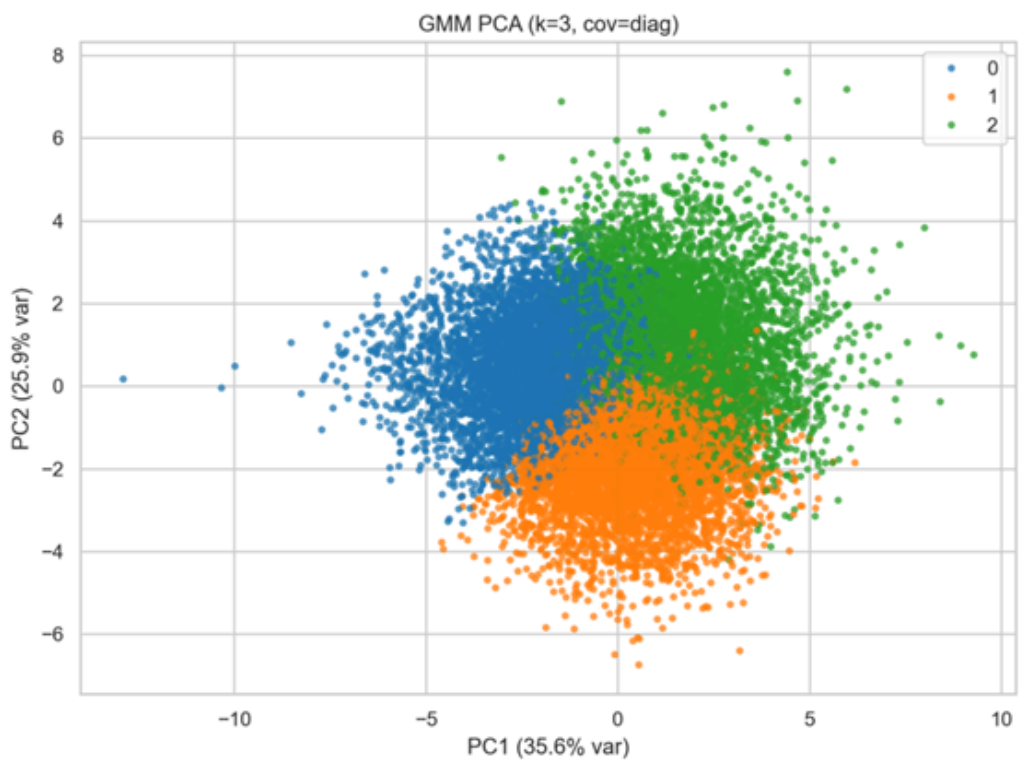

Figure 10. GMM components in PCA space

6.5 Sensitivity and Robustness

We assess robustness along three orthogonal axes, each targeting a distinct threat to validity:

Seed stability. We rerun K-Means (K=2, engineered features) with ten seeds (10–19). The mean Adjusted Rand Index (ARI) between all seed pairs is 0.996, indicating near-perfect partition agreement and confirming that the solution is not an artifact of initialization. This exceptionally high stability is consistent with K=2 partitions in high-sample-size regimes, where the two-cluster boundary is well-determined by the data geometry.

Subsample stability. We draw 12 random 80% subsamples and cluster each independently. The mean ARI across subsample-based comparisons is 0.970, showing that results are not driven by a small subset of influential observations. The modest variance across subsamples ($\sigma_ARI < 0.01$) further confirms structural robustness.

Proxy-weighting sensitivity. We perturb the N_proxy formula by varying the Openness weight from 0.5 to 1.0 (while adjusting the complementary Extraversion weight accordingly). Across all three variants the partition remains structurally consistent: the minimum ARI relative to the baseline is 0.838 (pure-Openness variant) and 0.885 (equal-weight variant), while Silhouette scores remain within 0.01 of the baseline (range 0.195–0.203). This demonstrates that cluster structure is robust to reasonable proxy-weighting choices and is not an artifact of specific formula calibration.

7. Discussion

7.1 Supervised Learning

Part A reinforced that data quality and preprocessing decisions have a greater impact on model performance than algorithm choice. The majority of analytical effort was spent before any model was trained — harmonising survey columns across nine years, identifying and correctly treating two fundamentally different types of missingness, and constructing a valid target variable from indirect signal. A key lesson was that feature engineering errors are not always visible at training time: the leakage proxy failure produced inflated metrics that appeared legitimate until feature importance inspection revealed an implausibly dominant feature, reinforcing interpretability tools as a routine evaluation step.

What surprised us: Geographic features were nearly as predictive as clinical comorbidity features — geographic ablation caused −0.076 recall drop vs −0.091 for comorbidity. We expected the workplace environment and MH disclosure behaviour to dominate. The sensitivity result was also counterintuitive: max_leaf_nodes=10 being optimal showed that with small minority class samples, aggressive pruning is a feature rather than a limitation.

Challenges: (1) Column and response harmonisation across nine survey years required FuzzyWuzzy fuzzy string matching with qualitative verification of each grouping. (2) Differentiating MNAR from MAR/MCAR required a year-stratified detection function — without this distinction, imputing MNAR values would fabricate answers to questions never posed, introducing year-confounded signal.

Extensions: A full factorial hyperparameter search to detect parameter interactions; ensemble methods (Random Forest, Gradient Boosting) to resolve the leaf resolution limit; region-stratified calibration to address non-American underprediction; ADHD-specific symptom features (inattention scales, age of onset, medication type) to replace the geographic proxy with individual-level clinical signal.

7.2 Unsupervised Learning (Part B)

Part B demonstrates that interpretable personality profiles can be recovered from Big Five trait data augmented with theory-driven proxy features, even though internal metrics indicate clusters are not sharply separated — consistent with the continuous, dimensional nature of personality variation. In this report, K=2 is used as the primary macro-archetype split, while K=3 is treated as a finer-grained refinement rather than a contradictory result.

What surprised us. Despite modest Silhouette values (≈0.20), the two-cluster K-Means solution produces clearly interpretable profiles that separate along a “high-trait/low-impulsivity” vs. “low-trait/high-impulsivity” axis. The K=3 solutions (K-Means and GMM) further reveal sub-profiles within the broader lower-trait group, differing primarily on J_proxy and T_proxy, hinting at finer-grained personality structure that a purely hard two-way split cannot resolve. This pattern parallels Gerlach et al.’s (2018) observation that personality types can emerge as distributional modes even when underlying traits are continuous.

Challenges encountered. (1) Proxy subjectivity: the MBTI-style proxy weights are theory-informed but not empirically calibrated from an independent mapping study. We mitigate this through the sensitivity analysis in Section 6.5, which demonstrates that reasonable weight perturbations do not materially alter cluster structure. (2) Evaluating unsupervised models without ground truth requires relying on internal metrics and interpretability rather

than classification accuracy — a fundamental limitation shared by all unsupervised personality research. (3) QC threshold selection (straight-lining ≥0.90, reverse-consistency >2.2) removed only 165 of 19,719 rows (0.84%); moderate tightening barely changes cluster assignments, supporting our chosen thresholds.

How Part B could be extended. Future work could incorporate density-based methods (HDBSCAN) or Spectral Clustering to capture non-convex cluster shapes; apply external validation using demographic or behavioral correlates (occupation, self-reported well-being); use Gower distance for mixed continuous–categorical features; conduct subgroup analyses (age bands, gender groups) with minimum cell-size thresholds ($n \geq 50$); calibrate proxy weights against an independent MBTI-labelled sample; and build an interactive dashboard for real-time profile exploration.

7.3 Cross-Part Triangulation

Taken together, Part A and Part B illuminate complementary facets of neurodiversity. Part A shows that ADHD signals in the workplace are best predicted by comorbidity patterns and geographic context rather than job-role or demographic features alone, while Part B reveals that personality trait space naturally separates into a "structured/conscientious" archetype and an "impulsive/exploration-oriented" archetype. The Part B "low-Conscientiousness, high-Openness" cluster aligns with personality profiles frequently associated with ADHD in clinical literature (Jacobsson et al., 2021), suggesting that the two analyses converge on a consistent portrait of neurodivergent tendencies—detectable through both supervised prediction from workplace surveys and unsupervised structure discovery from psychometric inventories. This convergence strengthens the overall project narrative: neurodiversity manifests in both environmental context (Part A) and intrinsic trait architecture (Part B), and understanding both perspectives is necessary for a complete picture.

8. Ethical Considerations

8.1 Part A

First, there is a risk of **privacy invasion**, especially if sensitive mental health or comorbidity features are misused beyond screening support purposes. Second, the model may amplify **geographic or cultural bias**, as seen in the continent-related signal, potentially reinforcing unequal diagnosis of certain groups of individuals. Third, false positives or false negatives could lead to inappropriate labeling or missed support, and it might require human effort to verify the results. These risks can be mitigated through strict data governance, transparent communication and disclosure that predictions reflect survey-derived signals rather than diagnoses, bias audits across subgroups, and limiting model use to voluntary, supportive screening rather than employment decision-making.

8.2 Part B

Labelling and reification risk. Assigning individuals to named clusters may encourage downstream users to treat statistical groupings as fixed personality types, potentially informing hiring or screening decisions the data cannot support.

Cultural and demographic bias. The Big Five model originates from English-language lexical studies and may not capture personality variation equally across cultures. The dataset is skewed toward English-speaking, internet-active respondents; cluster profiles should not be generalised without replication.

Privacy and re-identification risk. Although the dataset contains no direct identifiers, combinations of quasi-identifiers (country, age, gender, rare personality profiles) could theoretically narrow the anonymity set. Row-level cluster assignments are kept private; only aggregate profiles are reported.

9. Statement of Work

Zhiqi (Jenny) Zhu	Ki Yip (Keith) Chan	Yijing (Frank) Wang
Part B data processing: raw IPIP-FFM scoring, reverse-coding, quality control, z-standardisation, Part B data pipeline, report write up	Part A supervised learning: data merge, MNAR encoding, feature engineering, model training (Phase 1 & 2), sensitivity analysis, learning curve, feature importance, ablation, failure analysis, SHAP, report write up	Part B unsupervised learning: MBTI/Jung-style proxy engineering, clustering (KMeans/GMM), stability and sensitivity checks, visualisations, report write up.

Appendix A - Literature review

[1] Mumu, S. M., Hoque, H., & Sakib, N. (2022). *A classified mental health disorder (ADHD) dataset based on ensemble machine learning from social media platforms*. Ahsanullah University of Science and Technology.

[2] Kim, W. P., et al. (2023). *Machine learning–based prediction of ADHD and sleep problems with wearable data*. JAMA Network Open.

[3] Saar-Tsechansky, M. & Provost, F. (2007). **Handling Missing Values when Applying Classification Models.** *Journal of Machine Learning Research*, 8, 1623–1657. Accessed from February 12, 2026

[4] Costa, P. T., & McCrae, R. R. (1992). NEO-PI-R Professional Manual. Odessa, FL: Psychological Assessment Resources.

[5] Furnham, A. (1996). The big five versus the big four: the relationship between the Myers–Briggs Type Indicator (MBTI) and NEO-PI five factor model of personality. Personality and Individual Differences, 21(2), 303–307.

[6] Gerlach, M., Farb, B., Revelle, W., & Amaral, L. A. N. (2018). A robust data-driven approach identifies four personality types across four large data sets. Nature Human Behaviour, 2(10), 735–742.

[7] Jacobsson, P., Hopwood, C. J., Söderpalm, B., & Nilsson, T. (2021). Adult ADHD and emerging models of maladaptive personality: a meta-analytic review. BMC Psychiatry, 21(1), 282.

[8] Kim, W. P., et al. (2023). Machine learning–based prediction of ADHD and sleep problems with wearable data. JAMA Network Open.

[9] McCrae, R. R., & Costa, P. T. (1989). Reinterpreting the Myers–Briggs Type Indicator from the perspective of the five-factor model of personality. Journal of Personality.

[10] Mostafij Mumu, S., Hoque, H., & Sakib, N. (2022). A Classified Mental Health Disorder (ADHD) Dataset based on Ensemble Machine Learning from Social Media Platforms.

[11] Rasheed, F., Khan, S., Waqas, S. M., Khan, R. A., & Arif, S. (2023). Survey Dataset on Mental Health in Tech Professionals from Open Sourcing Mental Health Surveys (2017–2021). SSRN.

[12] Saar-Tsechansky, M. & Provost, F. (2007). Handling Missing Values when Applying Classification Models. Journal of Machine Learning Research, 8, 1623–1657.

[13] Van Ness, M., et al. (2023). The Missing Indicator Method: From Low to High Dimensions. Proceedings of the 2023 ACM Conference on Management of Data

Appendix B - Visualization reference

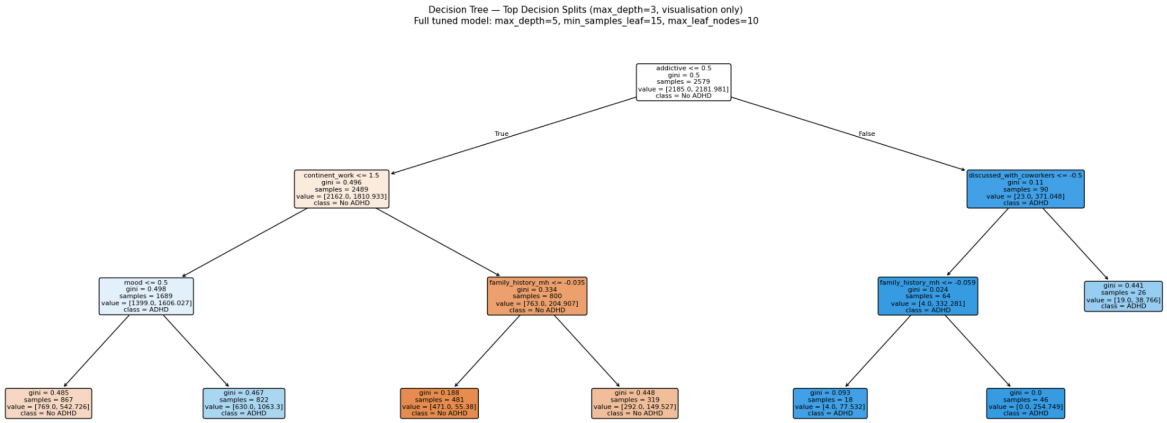

Figure B1: Tree diagram from Decision tree classifier model (Part A)

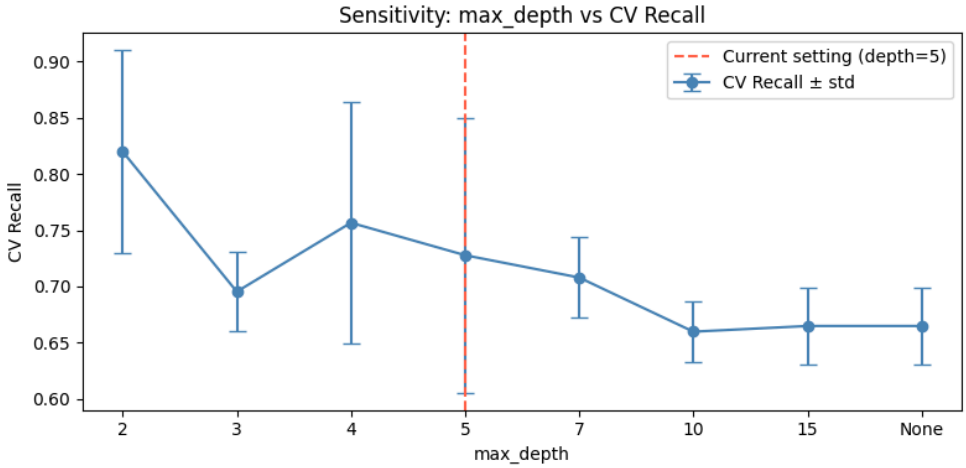

Figure B2: Sensitivity analysis results on max_depth (Part A)

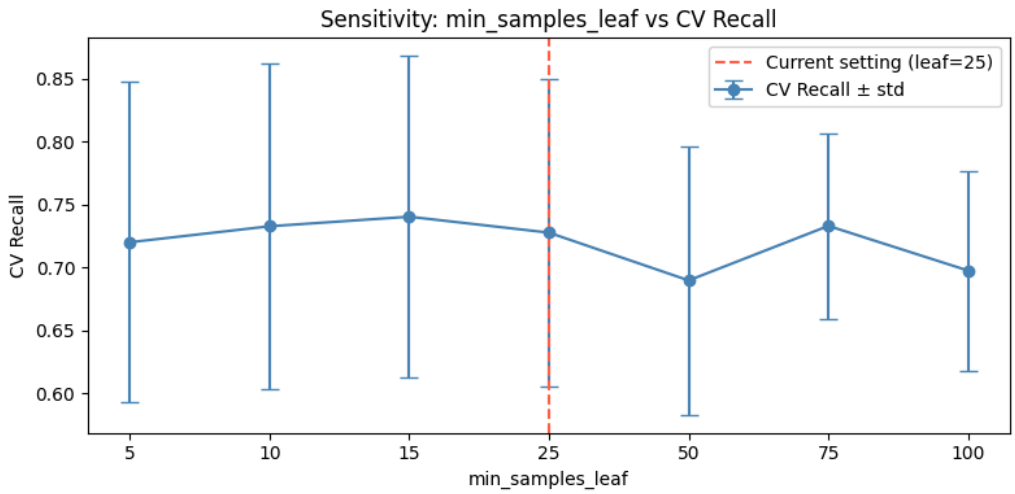

Figure B3: Sensitivity analysis results on min_samples_leaf (Part A)

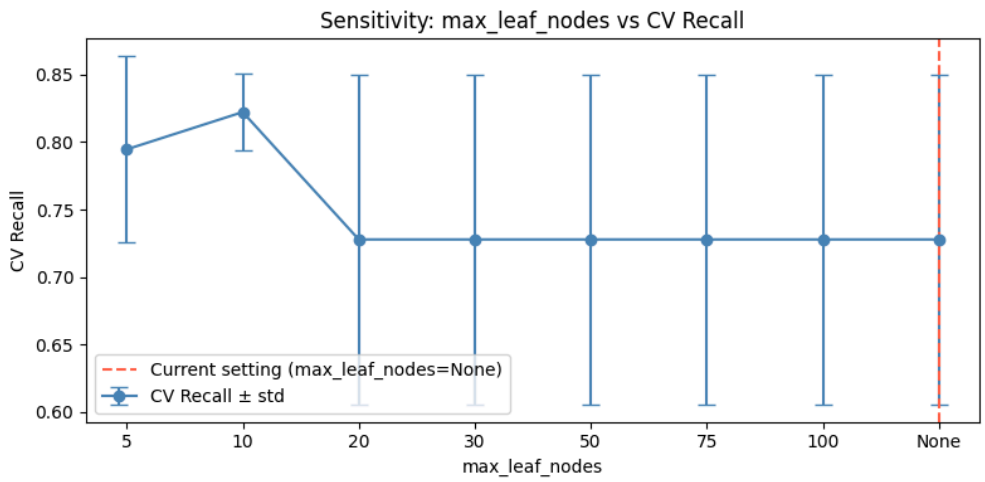

Figure B4: Sensitivity analysis results on max_leaf_nodes (Part A)

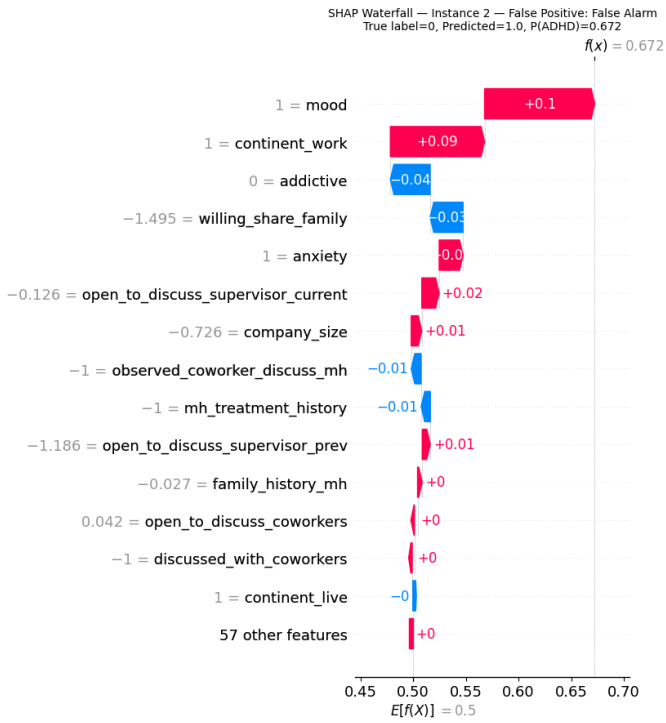

Figure B5: SHAP Waterfall plot – False Positive (non-ADHD flagged as ADHD)

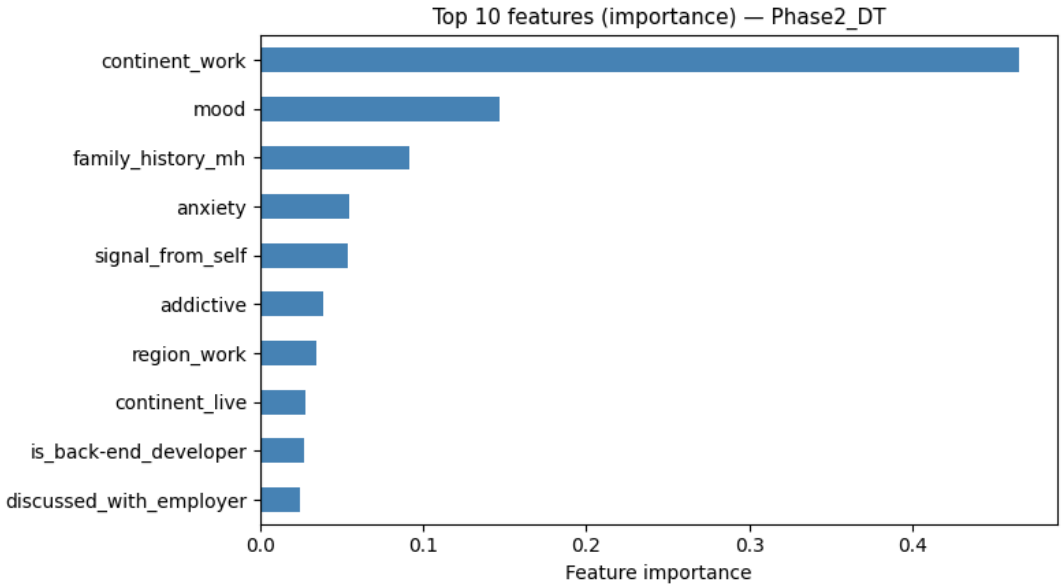

Figure B6: Failure analysis instance 3 – continent_work intensify due to 2014 data

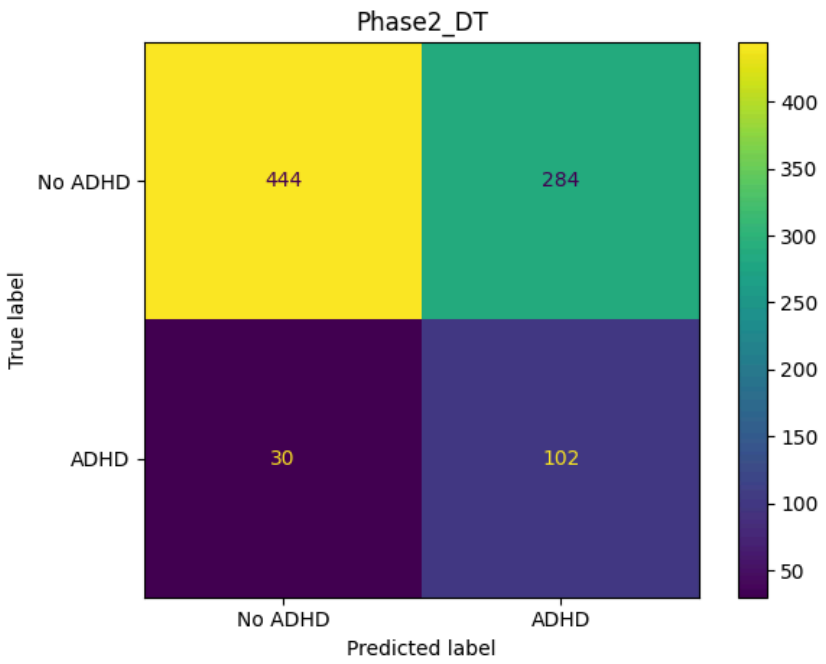

FIgure B7: Confusion matrix for Phase 2 Decision tree (before sensitivity analysis)

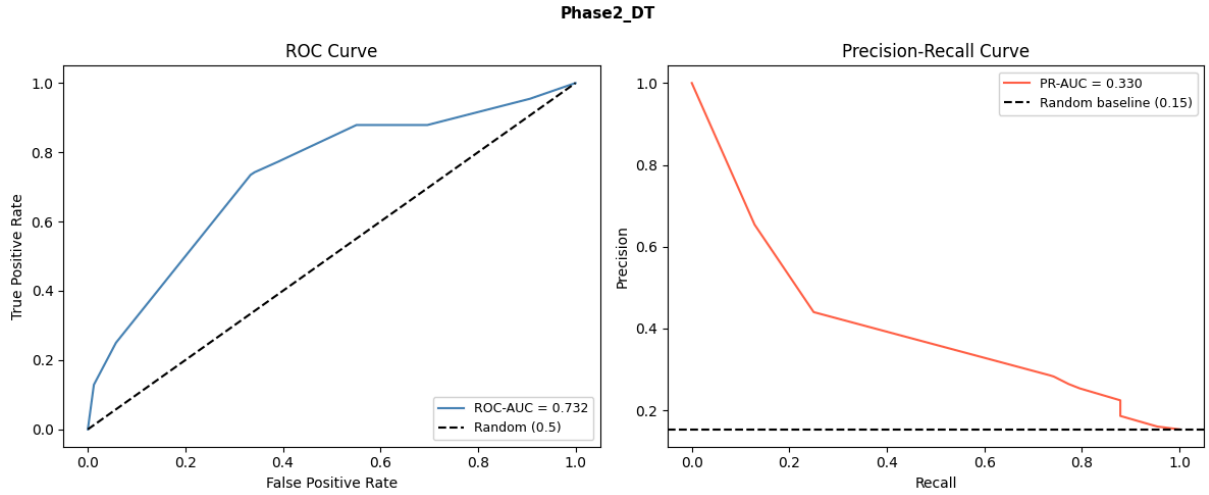

Figure B8: ROC-AUC (left) & PR-AUC (right) for phase 2 decision tree classifier (before sensitivity analysis)

Appendix C — Part B Data Schema

Data Source: http://openpsychometrics.org/_rawdata/ — Dataset: data.csv (tab-separated)

Table C1. Raw data columns.

Column(s)	Description	Type
race, age, engnat, gender, hand, source	Demographic variables	int
country	Country/region code	text
E1–E10	Extraversion items (Likert 1–5, includes reverse-keyed)	int
N1–N10	Neuroticism items	int
A1–A10	Agreeableness items	int

C1–C10	Conscientiousness items	int
O1–O10	Openness items	int

Table C2. Quality control rules.

Rule	Operationalization	Threshold
Straight-lining	Max single-choice proportion across 50 items	≥ 0.90
Reverse-consistency	Average mismatch score across paired items	> 2.2

Table C3. Dataset statistics.

Metric	Value
Raw rows	19,719
Raw columns	57
Rows removed by QC	165
Final rows analyzed	19,554

Appendix D — Part A Complete Feature List

The following table lists all final features used in the Part A supervised classification pipeline after preprocessing, harmonisation, and feature engineering. Features were derived from the pooled OSMI Mental Health in Tech Survey (2014–2023). Categorical features were one-hot or label-encoded; MNAR sentinel features have companion missingness indicator flags.

Table D1. Part A features — Demographics & Employment.

Feature	Description	Type
Age	Respondent age (clipped 16–100)	float
self_employed	Self-employment status	float
company_size	Employer company size category	int
tech_current	Currently works in tech	int
tech_previous	Previously worked in tech	float
tech_role	Current role is technical	float
job_spec	Job specialization category	cat
remote_work	Remote work status	int
prev_employer	Has previous employer	float
year	Survey year (2014–2023)	int
race_grouped	Standardized race category (7 groups + Missing)	cat
continent_live	Continent of residence (5 regions)	cat
continent_work	Continent of work (5 regions)	cat
region_live	US Census region (if applicable)	cat

region_work	US Census work region (if applicable)	cat

Table D2. Part A features — Mental Health & Comorbidity.

Feature	Description	Type
adhd	Self-reported ADHD diagnosis mention	int
anxiety	Anxiety disorder indicator	int
mood	Mood disorder indicator	int
eating	Eating disorder indicator	int
ocd	OCD indicator	int
ptsd	PTSD indicator	int
personality	Personality disorder indicator	int
psychotic	Psychotic disorder indicator	int
dissociative	Dissociative disorder indicator	int
addictive	Addictive disorder indicator	int
substance_use	Substance use disorder indicator	int
stress_response	Stress-response disorder indicator	int

mh_treatment_history	History of MH treatment	int
family_history_mh	Family history of mental health conditions	int
current_mh	Current mental health status	float
past_mh	Past mental health status	float
diagnosed_mh	Formally diagnosed MH condition	float
signal_from_clinical	Clinical signal derived flag	int
other_disorder	Other disorder indicator	int
signal_from_self	Self-report signal derived flag	int

Table D3. Part A features — Workplace Disclosure & Attitudes.

Feature	Description	Type
work_mh_benefits_current	Employer provides MH benefits	float
open_to_discuss_coworkers	Openness to discuss MH with coworkers	int
discussed_with_coworkers	Has discussed MH with coworkers	float
discussed_with_employer	Has discussed MH with employer	float
open_to_discuss_supervisor_current	Open to discuss with current supervisor	int

open_to_discuss_supervisor_prev	Open to discuss with previous supervisor	float
willing_share_family	Willing to share MH with family/friends	float
mh_share_interview	Would share MH in interview	int
ph_share_interview	Would share physical health in interview	int
discuss_mh_with_prev_coworkers	Discussed MH with previous coworkers	float
discuss_mh_with_prev_employer	Discussed MH with previous employer	float
disclose_mh_clients_business_current	Disclosed MH to clients/business contacts	float
disclose_mh_coworkers_current	Disclosed MH to current coworkers	float
observed_coworker_discuss_mh	Observed coworker discussing MH	float
past_disclose_mh_coworker	Past disclosure to coworker	float
vicarious_stigma_effect	Vicarious stigma experience	float
openly_identified_mh_at_work	Openly identified as having MH condition	float
willing_interview_followup	Willing to follow up in interview	float
prev_coworker_discussed_mh_with_you	Previous coworker discussed their MH	float

discuss_employer_neg_conseq	Negative consequences from employer discussion	int
mental_vs_physical_comfort_prev_employer	Comfort discussing MH vs PH (prev employer)	float
mental_vs_physical_comfort_coworker	Comfort discussing MH vs PH (coworker)	float
industry_support_mh	Perceived industry MH support	float

Target variable: adhd_signal (binary 1/0) — constructed from self-reported ADHD diagnosis mentions, explicitly treated as a survey-derived signal rather than a clinical diagnosis. Approximately 15% positive class.

Appendix E — Part B Complete Feature List

The following table lists all features used in Part B clustering, including raw Big Five trait scores, z-standardized scores, engineered MBTI-style proxies, composite indices, and output cluster labels.

Table E1. Part B complete feature inventory.

Feature	Definition / Description	Type
O_score, C_score, E_score, A_score, N_score	Mean trait scores after reverse coding	float
z_O, z_C, z_E, z_A, z_N	z-score standardized trait scores	float
N_proxy	Intuition proxy: 0.7×z_O + 0.3×z_E	float
J_proxy	Judging proxy: 0.8×z_C + 0.2×z_A	float
T_proxy	Thinking proxy: 0.6×z_C − 0.4×z_A	float
Impulsivity_Index	−z_C + z_N	float

Emotional_Reactivity	z_N − 0.3×z_A	float
Creative_Divergence	z_O − 0.2×z_C	float
Social_Adaptability	z_E + 0.5×z_A − 0.3×z_N	float
cluster_kmeans	K-Means cluster label (K=2)	int
cluster_gmm	GMM component label (K=3, diag)	int
cluster_best	Selected best-model cluster label	int

Appendix F — AIA-Aligned Ethics Checklist (Part B)

The following checklist is aligned to the Canadian Algorithmic Impact Assessment (AIA) framework structure. It is not the official AIA tool output but uses its categories to organize our ethical review of the Part B unsupervised clustering system.

Table F1. AIA-aligned assessment summary.

Topic	Assessment	Evidence
System type	Unsupervised clustering for exploratory archetype discovery	Part B scope
High-stakes decision use	No (explicitly out of scope)	Section 8
Uses personal information	Yes (psychometric responses; demographics present)	Minimize exports
Interpretability	Moderate (cluster profiles + proxy diagnostics)	Section 6.4
Stability checks performed	Yes	Seed ARI=0.996, Subsample ARI=0.970

Bias/fairness evaluation	Not implemented (requires n≥50 per subgroup)	Future work
Privacy controls	No re-identification; row-level outputs kept private	Policy + workflow

Table F2. Re-identification threat model.

Step	Demonstration (conceptual)	Performed?
1	Narrow records using quasi-identifiers (country, age, gender)	No
2	Intersect with rare/extreme personality profiles	No
3	Cross-reference external public sources	No
4	Infer identity or sensitive personal context	No

Table F3. Privacy mitigations.

Mitigation	Status	Notes
No re-identification attempts	Stated	Section 8
Aggregate reporting (cluster-level)	Implemented	Section 6.4
Do not publish row-level assignments	Required	cluster_assignments.csv kept private
Minimum cell-size thresholds for subgroups	Not implemented	Add before any subgroup appendix